Event analysis: ion production and diffusion

NAUSICAA

This checkpoint implements Petit's per-hit rounding and per-ion Gaussian Monte Carlo histogram, with explicit RNG inputs, Float32 physical arrays, Int64 counts, saved edges and a charge ledger. The cumulative Julia suite passes 1953 assertions. The pinned replay fixture agrees with Petit's original Float64 diffusion routine away from bin boundaries. Geometry, conservation, stochastic moments, finite-window losses, exact edges and reproducibility have separate tests.

Model and numerical conventions

The mean pair-creation energy is 21.9 eV. Counts are rounded independently per hit, with ties to even. The model uses a common drift distance per event, sampled uniformly from the configured interval. The thermal-width formula retains Petit's $k_B/e = 8.6173 \times 10^{-5}$ V/K, at 293.15 K and 200 V/cm. The baseline uses the same width on all axes. The zero-longitudinal-width comparison is supported. Coordinates remain in the source frame.

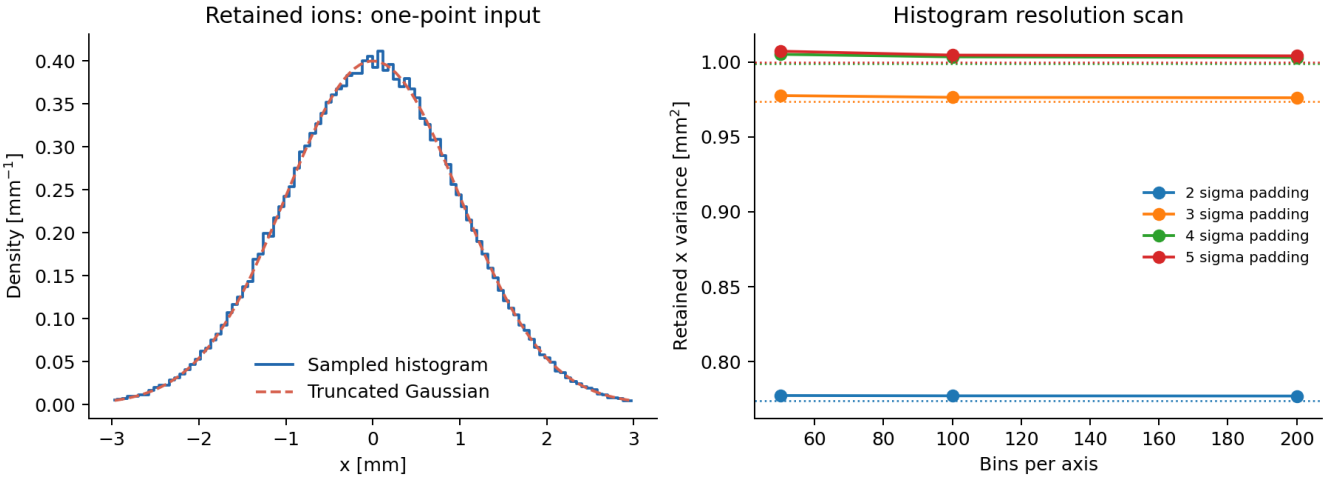
Histogram bounds span the hit extent plus the configured sigma padding. Internal bins are half open; the final upper edge is included. A coincident axis with zero diffusion uses one bin centred at that coordinate and adjacent representable Float32 edges. Every ion consumes three Gaussian draws. Outside counts remain in the ledger; retained counts are never renormalised. Int64 counts satisfy $N_{\text{sampled}} = N_{\text{retained}} + N_{\text{outside}}$ exactly.

Full-sample ion production

Sample	Events	Mean rounding [ions]	Max rounding [%]	Max arithmetic error
Double beta	8995	0.60	0.0172	2.57e-07
Electron background	8483	0.56	0.0172	3.27e-07

Rounding means sampled minus expected ions. The maximum rounding column uses the absolute event-relative difference. Arithmetic error compares Float32 ion expectations with an independent Float64 E/W sum; it is a relative error.

Diffusion checks



The point-source example has sigma=1 mm. The right panel compares retained histogram-centre variance with the analytic truncated-Gaussian variance (dotted lines). The same Gaussian draws are replayed at every bin count and padding. Increasing padding recovers the Gaussian tails; changing bin count preserves the retained count for fixed bounds. Finite-sample fluctuations remain after the histogram resolution converges.

Padding [sigma]	Sampled ions	Outside ions	Analytic outside [%]
2	200000	26043	13.038
3	200000	1615	0.80775
4	200000	32	0.019002
5	200000	0	0.00017199

Sample	Cases	Mean diffusion [ms]	Max outside [%]
Double beta	20	6.55	0.0321
Electron background	20	5.99	0.0205

The real-event checks cover events spread across each file, at the configured fixed drift distances and one sampled distance per event. Timings include histogram allocation and output construction after compilation; measured values depend on the machine. The HDF5 also records allocated bytes and seeds. These checks establish the diffusion stage. Selection efficiencies and pad histories belong to the next integration checkpoint.

Reproduction

```
make validate-event-step2 PYTHON=/path/to/python3.11-or-newer
# Regenerate presentation from saved HDF5:
make report-event-step2 PYTHON=/path/to/python3.11-or-newer
```

Source hashes, resolved inputs and measured tables are saved in:
`results/event_analysis/step2_validation.h5`.

The HTML review contains the implemented signatures and source docstrings.

Configuration used

```
# Ion-production/diffusion checkpoint. Detector constants come from the SI card.
[inputs]
detector_card = "config/detector.toml"
samples = ["events/0nubb_hitdf.h5", "events/electron_0nubb_energy_hitdf.h5"]

[ionization]
ionisation_energy_eV = 21.9
# Same adopted nominal W as the initial ion_tracks analysis.

[drift]
min_cm = 10.0
max_cm = 150.0

[diffusion]
longitudinal = "thermal" # "zero" is the longitudinal comparison
nbins = 100
nsigma = 3.0

[graph]
# Approved settings for the next integration stage.
method = "kNN"
k = 10
min_ions_per_bin = 10
bin_sigma_scale = 3.0
edge_sigma_scale = 6.0

[random]
seed = 2026
# Validation uses explicit per-case seeds, stored in its HDF5.

[validation]
events_per_sample = 5
synthetic_ions = 200000
drift_cm = [10.0, 80.0, 150.0]
bin_counts = [50, 100, 200]
padding_sigma = [2.0, 3.0, 4.0, 5.0]
```